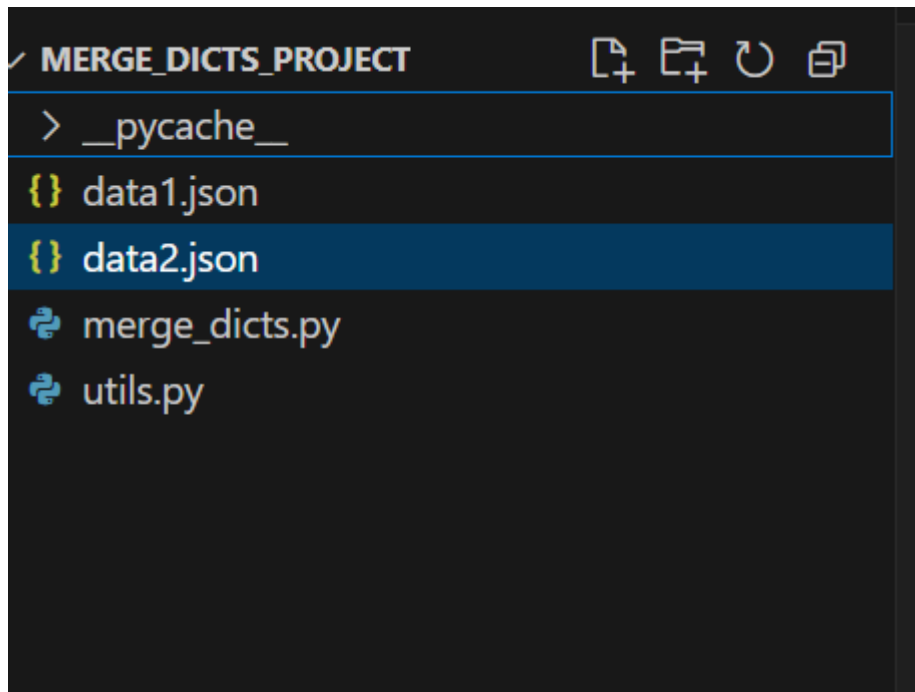


//Q1

//<https://github.com/AkashChand6n/Devops-restest.git>

//Python project combining values for keys that appear in both dictionaries.



```

utils.py
1  # utils.py
2  # Author: Akash Chandran
3  # Description: Helper functions for dictionary merging.
4
5  import json
6
7  def load_json(file_path):
8      """
9      Loads a JSON file and returns the data as a dictionary.
10     Args:
11         file_path (str): Path to the JSON file.
12     Returns:
13         dict: Parsed JSON data as a dictionary.
14     """
15     try:
16         with open(file_path, 'r') as file:
17             return json.load(file)
18     except FileNotFoundError:
19         print(f"Error: File '{file_path}' not found.")
20         return {}
21     except json.JSONDecodeError:
22         print(f"Error: Failed to decode JSON from '{file_path}'.")
23         return {}
24     except Exception as e:
25         print(f"An unexpected error occurred while loading '{file_path}': {e}")
26         return {}
27
28 def validate_dict(data):
29     """

```

```

def validate_dict(data):
    """
    Validates if the data is a dictionary.
    Args:
        data: The data to validate.
    Returns:
        bool: True if data is a dictionary, False otherwise.
    """
    if isinstance(data, dict):
        return True
    print("Error: The provided data is not a dictionary.")
    return False

```

```

# utils.py

# Author: Akash Chandran

# Description: Helper functions for dictionary merging.

import json

```

```

def load_json(file_path):
    """
    Loads a JSON file and returns the data as a dictionary.
    Args:
        file_path (str): Path to the JSON file.
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    if isinstance(data, dict):
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print("Error: The provided data is not a dictionary.")

return False
```

```
merge_dicts.py
1  # merge_dicts.py
2  # Author: Akash Chandran
3  # Description: Merges two dictionaries, combining values for common keys.
4
5  from utils import load_json, validate_dict
6
7  def merge_dicts(dict1, dict2):
8      """
9      Merges two dictionaries, combining values for keys that appear in both dictionaries.
10     Args:
11         dict1 (dict): The first dictionary.
12         dict2 (dict): The second dictionary.
13     Returns:
14         dict: A dictionary that combines values from both input dictionaries.
15     """
16     try:
17         if not validate_dict(dict1) or not validate_dict(dict2):
18             return {}
19
20         # Combine dictionaries
21         merged_dict = dict1.copy() # Copy first dictionary to avoid modifying the original
22         for key, value in dict2.items():
23             if key in merged_dict:
24                 # Combine values (can be adjusted based on how values should be merged)
25                 merged_dict[key] += value
26             else:
27                 merged_dict[key] = value
28
29         return merged_dict
```

```

9         return merged_dict
10    except Exception as e:
11        print(f"An unexpected error occurred while merging dictionaries: {e}")
12        return {}
13
14    if __name__ == "__main__":
15        # Example usage of loading JSON files and merging them
16        dict1 = load_json('data1.json')
17        dict2 = load_json('data2.json')
18
19        # Merge the dictionaries
20        merged_result = merge_dicts(dict1, dict2)
21
22        if merged_result:
23            print("Merged Dictionary:")
24            print(merged_result)
25        else:
26            print("Error: Could not merge dictionaries.")
27

```

merge_dicts.py

Author: Akash Chandran

Description: Merges two dictionaries, combining values for common keys.

```
from utils import load_json, validate_dict
```

```
def merge_dicts(dict1, dict2):
```

```
    """
```

```
    Merges two dictionaries, combining values for keys that appear
    in both dictionaries.
```

```
    Args:
```

```
        dict1 (dict): The first dictionary.
```

```
        dict2 (dict): The second dictionary.
```

```
    Returns:
```

```
        dict: A dictionary that combines values from both input
        dictionaries.
```

```
    """
```

```

try:
    if not validate_dict(dict1) or not validate_dict(dict2):
        return {}

    # Combine dictionaries

    merged_dict = dict1.copy() # Copy first dictionary to avoid
modifying the original
    for key, value in dict2.items():
        if key in merged_dict:
            # Combine values (can be adjusted based on how
values should be merged)
            merged_dict[key] += value
        else:
            merged_dict[key] = value

    return merged_dict
except Exception as e:
    print(f"An unexpected error occurred while merging
dictionaries: {e}")
    return {}

if __name__ == "__main__":
    # Example usage of loading JSON files and merging them
    dict1 = load_json('data1.json')
    dict2 = load_json('data2.json')

    # Merge the dictionaries
    merged_result = merge_dicts(dict1, dict2)

    if merged_result:
        print("Merged Dictionary:")
        print(merged_result)
    else:
        print("Error: Could not merge dictionaries.")

```

```
{ } data1.json > ...
```

```
1  {  
2      "a": 1,  
3      "b": 2,  
4      "c": 3  
5  }  
6
```

```
{ } data2.json > ...
```

```
1  {  
2      "b": 4,  
3      "c": 5,  
4      "d": 6  
5  }  
6
```

```
\Desktop\merge_dicts_project> python merge_dicts.py
```

```
C:\Users\290593\Desktop\merge_dicts_project> python merge_dicts.py
```

```
Merged Dictionary:
```

```
{'a': 1, 'b': 6, 'c': 8, 'd': 6}
```

```
Press any key to continue . . .
```